

Empirical and Molecular Formulas

Answer key with worked solutions

A. Understanding

1. Empirical vs molecular formula.

The **empirical** formula gives the simplest whole-number ratio of atoms; the **molecular** formula gives the actual number of each kind of atom in one molecule.

2. Empirical formulas.

(a) CH (b) CH₂ (c) HO (d) C₄H₉ (e) NO₂

3. What is wrong with rounding 1 : 1.50?

1.50 is not close to a whole number, so it must not be rounded. Multiply both values by 2: 2 : 3, giving A₂B₃.

B. Empirical formulas from percent composition

In a 100 g sample, each percentage becomes a mass in grams.

4. 85.63% C, 14.37% H.

C: $85.63 \text{ g} \div 12.01 \text{ g/mol} = 7.130 \text{ mol}$; H: $14.37 \text{ g} \div 1.008 \text{ g/mol} = 14.26 \text{ mol}$. Divide by 7.130 mol: C = 1.00, H = 2.00. Empirical formula CH₂.

5. 40.04% Ca, 12.00% C, 47.96% O.

Ca: $40.04 \text{ g} \div 40.08 \text{ g/mol} = 0.9990 \text{ mol}$; C: $12.00 \text{ g} \div 12.01 \text{ g/mol} = 0.9992 \text{ mol}$; O: $47.96 \text{ g} \div 16.00 \text{ g/mol} = 2.998 \text{ mol}$. Divide by 0.9990 mol: 1 : 1 : 3. Empirical formula CaCO₃.

6. 72.36% Fe, 27.64% O.

Fe: $72.36 \text{ g} \div 55.85 \text{ g/mol} = 1.296 \text{ mol}$; O: $27.64 \text{ g} \div 16.00 \text{ g/mol} = 1.728 \text{ mol}$. Divide by 1.296 mol: Fe = 1.000, O = 1.333. A ratio ending in .33 means multiply by 3: 3 : 4. Empirical formula Fe₃O₄.

C. Molecular formulas

7. CH₂O, molar mass 60.05 g/mol.

Empirical formula mass = 12.01 g/mol + 2(1.008 g/mol) + 16.00 g/mol = 30.03 g/mol. $n = 60.05 \text{ g/mol} \div 30.03 \text{ g/mol} = 2.00$. Molecular formula C₂H₄O₂.

8. CH, molar mass 26.04 g/mol.

Empirical formula mass = 12.01 g/mol + 1.008 g/mol = 13.02 g/mol. $n = 26.04 \text{ g/mol} \div 13.02 \text{ g/mol} = 2.00$. Molecular formula C₂H₂.

D. Challenge

9. 30.43% N, molar mass 92.02 g/mol.

O = 100% – 30.43% = 69.57%. In 100 g: N: 30.43 g ÷ 14.01 g/mol = 2.172 mol; O: 69.57 g ÷ 16.00 g/mol = 4.348 mol. Ratio N : O = 1 : 2.00, so the empirical formula is NO₂ (46.01 g/mol).

$n = 92.02 \text{ g/mol} \div 46.01 \text{ g/mol} = 2.00$, so the molecular formula is N₂O₄.